

Full Name: _____
EEL 3135 (Spring 2026) – Lab #10

What to Submit on Canvas:

- M-File: Modified `ee13135_lab09_skeleton.m` file (initial template found on Canvas)
- PDF: A single published MATLAB PDF of your modified skeleton code

Submission Guidelines:

- **When creating new functions:**
 - Place them at the end of the skeleton M-file, unless otherwise specified
- **When plotting:**
 - Label all plots with `xlabel` and `ylabel` to describe axes and units (when appropriate)
- **When plotting time-domain signals:**
 - Choose a time axis that shows the primary characteristics of the filter or signal, unless otherwise specified
 - `stem` figures should have x-axes in samples and start at $n = 0$
 - `plot` figures should have x-axes in actual time (usually seconds) and start at $t = 0$
- **When plotting frequency-domain signals:**
 - Choose a frequency axis that shows the primary characteristics of the filter or signal, unless otherwise specified
 - Frequency axis should be in normalized radians and show a range from $-\pi$ to π , unless otherwise specified
- **When plotting images**
 - All images should use `colormap(gray)` to display images in grayscale
 - All images should use `axis image` to show images in an image orientation
- **When saving audio:**
 - Rescale vectors to have a range of values between -1 and 1 before passing them to `audiowrite` to make your .wav files, or the audio will distort.

Useful Functions to Know:

- `conv`, `fft`, `tic`, `toc`, DTFT (provided)

Question #1: (*Implementing the DFT*) In `ee13135_lab10_comment.m`, replace appropriate comments with the descriptions to help you in the lab.¹

Question #2: (*DFT and DTFT*) In this question, you create a DFT function and compare results with the DTFT function. Keep in mind that the DFT can be thought of as a sampled version of the DTFT. Consider the following discrete-time signal:

$$x[n] = [0.75 + \cos((\pi/20)n) + \cos((\pi/15)n) + \cos(\pi n + 2\pi/3)] [u[n] - u[n - 60]]$$

This signal is found in the skeleton code. We also provide `w_DTFT`, the DTFT frequencies to use for this exercise (we will plot the DTFT / DFT for 0 to 2π instead of $-\pi$ to π in this lab).

- (a) Create the `X = DFT(x)` function, which inputs a signal `x` and outputs its DFT `X` with the same length.

Hint 1: The DTFT (from previous labs) and length- N DFT have similar definitions:

$$\text{DTFT: } X(\hat{\omega}) = \sum_{n=0}^{N-1} x[n]e^{-j\hat{\omega}n} \quad , \quad \text{DFT: } X[k] = \sum_{n=0}^{N-1} x[n]e^{-j(\frac{2\pi}{N})kn}$$

Hint 2: The `w` variable in the skeleton code's `DFT` function should be in normalized angular frequency $\hat{\omega}$ and can be determined by directly comparing the DFT and DTFT equations.

- (b) Plot the magnitude of the DTFT of $x[n]$. Be sure to label axes with proper units.
- (c) Plot the magnitude of the 60-length DFT of $x[n]$ on top of the DTFT. See the comment code on how to best plot the DFT on top of the DTFT plot. Note: `w_DFT` is the same as `w` in the DFT function.
- (d) Plot the magnitude of the 55-length DFT (i.e., remove the last 5 values from the original signal) of $x[n]$ on top of the DTFT.
- (e) Plot the magnitude of the 65-length DFT (i.e., add 5 zeros to the original signal) of $x[n]$ on top of the DTFT.
- (f) Plot the magnitude of the 200-length DFT of $x[n]$ on top of the DTFT.
- (g) **Answer in your comments:** Based on the last several questions, what is the relationship between the DTFT and the DFT? Under what conditions will the theoretical DTFT and DFT have the same result?

Question #3: (*DFT and IDFT*) In this question, you create the inverse DFT function and use it with the DFT function to perform convolution. Consider the following discrete-time signal:

$$x[n] = u[n - 30] - u[n - 65]$$

¹**Hint:** Run the code to understand it. You can / should use this MATLAB code in the remainder of the lab.

- (a) Create the `x = IDFT(X)` function, which inputs a DFT signal `X` and outputs its time-domain signal `x` with the same length.

Hint: The inverse DFT is defined as

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+j(\frac{2\pi}{N})kn}$$

- (b) Plot (for $0 \leq n \leq 99$) $x[n]$. Use the `stem` function.
- (c) Plot (for $0 \leq n \leq 99$) $y[n] = x[n] * x[n]$. Use the `stem` function.
- (d) Plot (for $0 \leq n \leq 99$) $\text{IDFT}\{\text{DFT}\{x[n]\}\text{DFT}\{x[n]\}\}$. Use $N = 100$ length DFT and IDFT functions. Use the `stem` function.
- (e) Plot (for $0 \leq n \leq 99$) $\text{IDFT}\{\text{DFT}\{x[n]\}\text{DFT}\{x[n]\}\}$. Use $N = 200$ length DFT and IDFT functions. Use the `stem` function.
- (f) **Answer in your comments:** What is the difference in the last two solutions? Why does the difference exist?

Question #4: (*DFT and FFT*)

This question will focus on computation time differences between DFT and FFT. Choose a song at least 3 minutes long for this problem and include it in your submission. Load it into MATLAB using `audioread`. Note that most audio files will be stereo, so you need to make sure that you only use one column of the audio data for this part of the lab (e.g., set `x = x(:,1);`). This site has a large archive of free music that you can choose from: <https://freemusicarchive.org/static>.

- (a) Plot the magnitude of the DFT of only the first 10000 samples of the audio and display (using the `disp` function) the output of `tic` and `toc` (read `help tic` and `help toc`) to measure the length of time it takes to compute the DFT, in seconds.
- (b) Plot the magnitude of the FFT of only the first 10000 samples of the audio and display (using the `disp` function) the output of `tic` and `toc` to measure the length of time it takes to compute the FFT, in seconds.
- (c) **Answer in your comments:** Are there differences in the magnitude results? If so, why?
- (d) **Answer in your comments:** How much faster is the FFT algorithm compared with the DFT in this scenario?
- (e) Display (using the `disp` function) the output of `tic` and `toc` to measure the length of time it takes to compute the FFT of the entire song, in seconds. Using DFT on an entire 3 minute audio (sampled at 44.1 kHz) should take approximately 500 hours, so let's not do that one.